Lecture 9 - Linearization and Derivative Rules

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10/08/2026

Post Exam Recap

1. Solutions to Exam 1 are up on Canvas, I'm happy to answer any questions about it if people want
2. Exams should be graded by tonight or tomorrow night, with curve details to follow after I release the scores
3. One important note so far: follow the instructions in the questions! Many people did not include reasoning for parts b and c of question 2; this will decrease your scores!

Quiz Time!

Let’s get into neat rows and spread out for the quiz!

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Just kidding :) no quiz today



Linearization

Why we care!

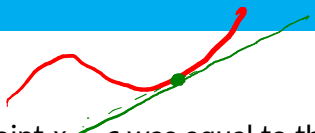
Best Linear Approximation



Recall that the *derivative* of a function $f(x)$ at a point $x = a$ was equal to the slope of the *best linear approximation* to $f(x)$ at $x = a$.

Linearization

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$$x \approx a \implies f(x) \approx f(a) + f'(a)(x - a). \tag{1}$$

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The linearization of a differentiable function $f(x)$ at a point $x = a$ is the best linear approximation to $f(x)$ at $x = a$ and is equal to

$$\mathscr{L}_{x=a}(x) = f(a) + f'(a)(x - a) \tag{2}$$

Using the Linearization to Approximate I

Using linearizations gives us *the strongest* ways approximate functions in a quick and accurate way.

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Example 1: The *logistic population growth* model is a mathematical description of how the size of a population $P(t)$ grows over some time period t with some growth factor k and some carrying capacity N. It assumes that:

$$\frac{dP}{dt} = kP(N - P) \tag{3}$$

Suppose that we have a population $P(t)$, where t is measured in days, which:

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What is our best estimate for the population size one year after we started measuring?

Using the Linearization to Approximate II

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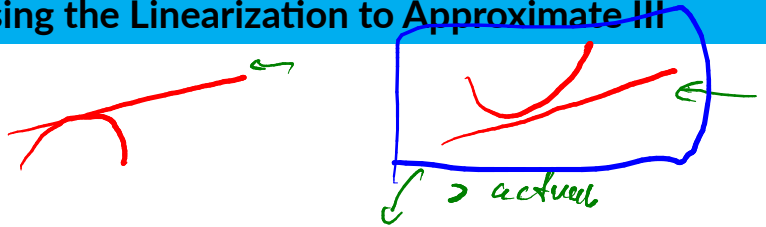
$$P(1) \approx P(0) + P'(0)(1 - 0) = 1000 + 1000 \cdot 1 = 2000 \tag{9}$$

Using the Linearization to Approximate III



Is this an over-approximation ($\mathscr{L}_{t=0}(1) > P(1)$) or an under-approximation ($\mathscr{L}_{t=0}(1) < P(1)$)?
What would we need to determine that? concavity! sign of 2nd deriv!

Using the Linearization to Approximate III

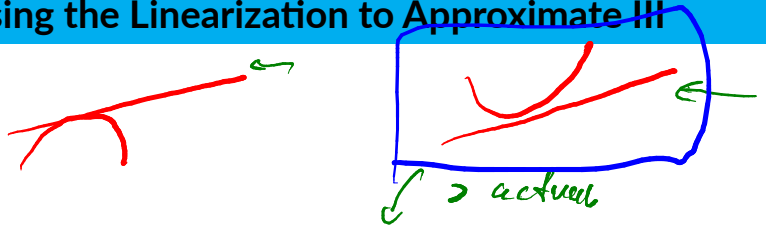


Is this an over-approximation ($\mathcal{L}_{t=0}(1) > P(1)$) or an under-approximation ($\mathcal{L}_{t=0}(1) < P(1)$)? What would we need to determine that? Suppose we know that $P''(0) = 900$. Does that mean our linearization gave an over-approximation or an under-approximation?

$P''(0) = 900 > 0$ $\quad \mathcal{L}_{x=0}(1) = 2000$

$0 \Rightarrow$ concave up

Using the Linearization to Approximate III

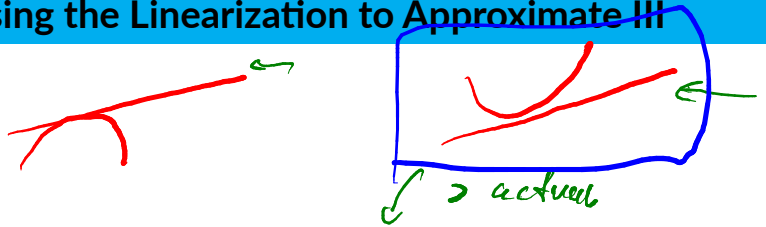


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Suppose we know that $P''(0) = 900$. Does that mean our linearization gave an over-approximation or an under-approximation?
Since $P''(0) > 0$, $P(t)$ is concave up: $d > 0$

$P''(0) = 900 < \mathcal{L}_{x=0}(1) = 2000$ down < 0

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Using the Linearization to Approximate III



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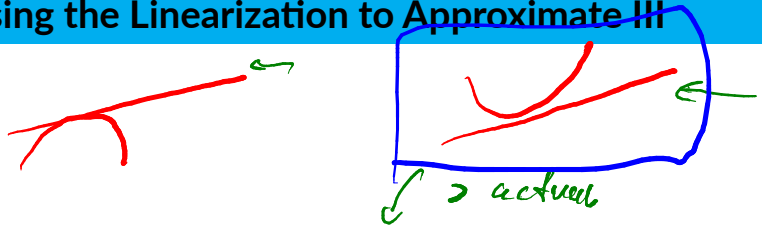
concave up: d > 0

down < 0

$P''(0) = 900 > 0 \Rightarrow$ concave up

$\mathcal{L}_{x=0}(1) = 2000$

Using the Linearization to Approximate III



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What would we need to determine that? concavity! sign of 2nd deriv!
Suppose we know that $P''(0) = 900$. Does that mean our linearization gave an over-approximation or an under-approximation?
Since $P''(0) > 0$, $P(t)$ is *concave-up* at $t = 0$, which means our approximation is an *under* approximation!
(Concave up => graph is above its tangent line)

concave up: $d > 0$ down < 0

$P''(0) = 900 > 0$ $\mathcal{L}_{x=0}(1) = 2000$

0 => concave up

Using the Linearization to Approximate IV

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1. Easy to define but

Using the Linearization to Approximate IV

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2. Hard to evaluate

like $f(x) = \sqrt{x}$, around known values.

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Using the Linearization to Approximate IV

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Example 2: Approximate $\sqrt{25.1}$.

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Approximation should always have us think of linearization: what do we need to linearize?

Using the Linearization to Approximate IV

$\mathcal{L}_{x=4}(x)$

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We need:

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Using the Linearization to Approximate IV

$\mathcal{L}_{x=a}(x) \doteq f(a)$

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We can also approximate functions that are: $\approx f(x)$

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We need:

1. An initial value $x = a$ around which to approximate where
2. We know $f(a)$ AND
3. We know $f'(a)$.

What is a value $a \approx 25.1$ where we know $\sqrt{a}$ and the derivative of the sqrt function at $x = a$?

Using the Linearization to Approximate V

We know that:

Using the Linearization to Approximate V

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Using the Linearization to Approximate V

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$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}} \tag{10}$$

so knowing $\frac{d}{dx}|_{x=a}\sqrt{x}$ is as hard as knowing $\sqrt{a}$. What should a be?

We should set $a = 25$ since $25 \approx 25.1$ AND $\sqrt{25} = 5$ is easy!

Then we get:

$$\sqrt{25.1} \approx \sqrt{25} + \frac{1}{2 \cdot \sqrt{25}}(25.1 - 25) \tag{11}$$

$$= 5 + \frac{1}{2 \cdot 5} \cdot (0.1) \tag{12}$$

$$= 5 + \frac{1}{10} \cdot 0.1 \tag{13}$$

$$= 5 + 0.01 = 5.01 \tag{14}$$

Using the Linearization to Approximate VI

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Is our approximation then an over or under approximation?
Well:

$$\frac{d^2}{dx^2}\Big|_{x=5}\sqrt{x} = \left(-\frac{1}{4\sqrt{x^3}}\right)\Big|_{x=5} \tag{16}$$

$$= -\frac{1}{4\sqrt{125}} < 0 \tag{17}$$

So this is an *over* approximation since we are concave down at $x = 5$.

Derivative Rules



Goodbye Limits!

Limit Definition is Hard

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$$= \frac{1}{2}\lim_{h\to 0}\frac{\sqrt{x^3+hx^2} - \sqrt{(x+h)^2\cdot x}}{hx(x+h)} \tag{21}$$

Limit Definition is Hard

How can we get that $\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{-1}{4\sqrt{x^3}}$?

Terrible!

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Limit Definition is Hard

How can we get that $\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{-1}{4\sqrt{x^3}}$?

Terrible!

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GROSS!!

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$$= \frac{-1}{4\sqrt{x^3}} \tag{27}$$

power rule

constant multiple rule

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- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
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- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

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$$(f(a) + f'(a)(x-a)) + (g(a)g'(a)(x-a)) = (f(a) + g(a)) + (f'(a) + g'(a))(x-a) \tag{29}$$

Constant Multiple Rule

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The derivative of a function multiplied by a constant is that constant multiplied by the derivative of the function:

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The factor must actually be a constant!

$$\frac{d}{dx}x^2 = 2x \neq x \cdot \frac{d}{dx}x = x \cdot 1 = x \tag{32}$$

Example

We can use this to compute derivatives of functions like:

$$f(x) = 5x^3 + 3x^2 + 2x + 5 \tag{33}$$

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$$= 15x^2 + 6x + 2 \tag{40}$$

Power Rule I

What about for more complicated polynomials

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$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

What about for more complicated polynomials e.g. what if the degree is > 3?

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Can we see a pattern?
Guess: $\frac{d}{dx}x^n = n \cdot x^{n-1}$.

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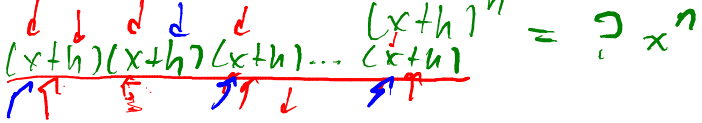
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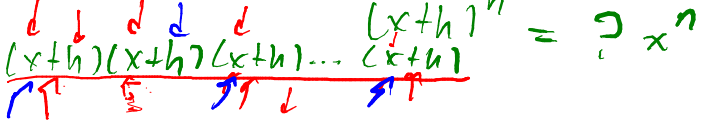
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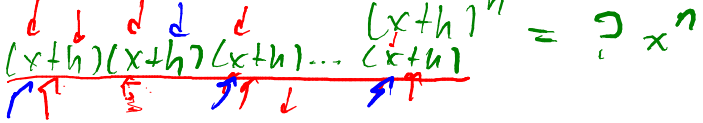
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So we know that $\frac{d}{dx}x^n = nx^{n-1}$!

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Power rule only applies to ***POWER FUNCTIONS*** i.e. $f(x) = x^c$ for some constant c.
It does NOT apply to *exponential functions* like $f(x) = 2^x$. They may look similar, but their derivatives are VASTLY different!

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it is just some constant multiple of the original function!

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What is the derivative of an exponential function?

$$\frac{d}{dx}c^x = \lim_{h\to 0}\frac{c^{x+h}-c^x}{h} \tag{50}$$

$$= \lim_{h\to 0} c^x \frac{c^h-1}{h} \tag{51}$$

$$= \left(\lim_{h\to 0}\frac{c^h-1}{h}\right)c^x \tag{52}$$

it is just some constant multiple of the original function!

What is the constant factor $\lim_{h\to 0}\frac{c^h-1}{h}$?

Exponential Rule II

Define a constant e so that $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$ (i.e. $\frac{d}{dx} e^x = e^x$). Then:

$$\lim_{h \to 0} \frac{c^h - 1}{h} =$$

Exponential Rule II

Define a constant e so that $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$ (i.e. $\frac{d}{dx} e^x = e^x$). Then:

$$\lim_{h \to 0} \frac{c^h - 1}{h} = \lim_{h \to 0} \frac{(e^{\ln(c)})^h - 1}{h} \tag{53}$$

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$$= \lim_{h \to 0} \frac{\ln(c)}{\ln(c)} \frac{e^{\ln(c)h} - 1}{h} \tag{55}$$

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$$= \lim_{h\to 0} \frac{\ln(c)}{\ln(c)} \frac{e^{\ln(c)h} - 1}{h} \tag{55}$$

$$= \ln(c) \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{\ln(c)h} \tag{56}$$

Exponential Rule II

Define a constant e so that $\lim_{h\to 0} \frac{e^h - 1}{h} = 1$ (i.e. $\frac{d}{dx} e^x = e^x$). Then:

$$\lim_{h\to 0} \frac{c^h - 1}{h} = \lim_{h\to 0} \frac{(e^{\ln(c)})^h - 1}{h} \tag{53}$$

$$= \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{h} \tag{54}$$

$$= \lim_{h\to 0} \frac{\ln(c)}{\ln(c)} \frac{e^{\ln(c)h} - 1}{h} \tag{55}$$

$$= \ln(c) \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{\ln(c)h} \qquad z = \ln(c)\,h \tag{56}$$

$$= \ln(c) \lim_{z\to 0} \frac{e^z - 1}{z} = \ln(c) \cdot 1 = \ln(c) \tag{57}$$

Exponential Rule III

This gives us:

Exponential Rule

The derivative of an exponential function, i.e. c^x for a constant c is:

$$\frac{d}{dx}c^x = \ln(c)c^x \tag{58}$$

Exponential Rule III

This gives us:

Exponential Rule

The derivative of an exponential function, i.e. c^x for a constant c is:

$$\frac{d}{dx}c^x = \ln(c)c^x \tag{58}$$

Exponential functions growing rapidly => it's derivative grows very large very quickly.
Very different behavior to that for power rule $\frac{d}{dx}x^c = cx^{c-1}$.

Exponential Rule IV

What is the n'th derivative of an exponential function c^x?

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Second derivative: $\frac{d}{dx}(\ln(c)c^x) = \ln(c)\frac{d}{dx}c^x = \ln(c)^2c^x$

Exponential Rule IV

What is the n'th derivative of an exponential function c^x?
First derivative: $\ln(c)c^x$.
Second derivative: $\frac{d}{dx}(\ln(c)c^x) = \ln(c)\frac{d}{dx}c^x = \ln(c)^2c^x$
Third derivative: $\ln(c)^3c^x$.

Exponential Rule IV

$$\frac{d^n}{dx^n} e^x = e^x$$

What is the n'th derivative of an exponential function c^x?

First derivative: $\ln(c)c^x$.

Second derivative: $\frac{d}{dx}(\ln(c)c^x) = \ln(c)\frac{d}{dx}c^x = \ln(c)^2 c^x$

Third derivative: $\ln(c)^3 c^x$.

...

n'th derivative: $\ln(c)^n c^x$.

$$\frac{d^n}{dx^n} c^x = \ln(c)^n c^x$$

Worksheet Time!

Last problem talks about product rule

ignore it.

Let's get into groups and work on the worksheet!

End of Class

That’s all folks!